

Programming Education in the Era of AI

Perspectives from Brazilian Higher Education

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Agenda

- Motivation
- Problem
- Background
- Related Work
- The Research Gap
- Research Proposal
- Study Design
- Contributions
- Limitations
- Conclusion

Motivation

- Increasing use of AI assistants in Software Engineering activities
- Based on LLMs, tools as ChatGPT and Gemini provide an interactive interface
- Higher education students in computing-related fields using LLMs in various programming scenarios
 - Learning concepts
 - Code generation for implementation tasks
- Local gap
 - Need to understand these dynamics specifically in Brazil

Problem

- Several challenges and risks that must be considered
- From the students' perspective, the main difficulties are:
 - Crafting effective prompts
 - Obtaining accurate and relevant answers when interacting with LLMs
- From the instructors' perspective:
 - Negative correlation between grades and the use of LLMs
 - LLMs may negatively affect students' ability to solve programming problems independently
 - Concerns regarding learning, over-reliance and plagiarism

Background

The Brazilian context

- GenAI is widely used in Brazil, as illustrated by ChatGPT
 - 140 million messages from the country every day
 - 15% related to learning
 - 6% related to programming, data science, and mathematics
- Diverse educational context and sociocultural landscape
 - According to INEP, in 2024:
 - More than 10.2 million enrollments in undergraduate programs
 - 50.7% of students in distance education, across public and private institutions

Related Work

- Raihan et al. [2025]
 - Noted student frustration with prompts and instructor concerns about over-reliance, urging curriculum updates
- Geng et al. [2026]
 - Examined "vibe coding," finding students rely on prompts for testing and debugging, though experienced coders write better prompts
- Ferino et al. [2026]
 - Found that novice developers commonly use LLMs for development tasks such as code generation, but highlighted gaps regarding vibe coding practices and the risk that learners may copy code without developing deeper understanding.

Related Work

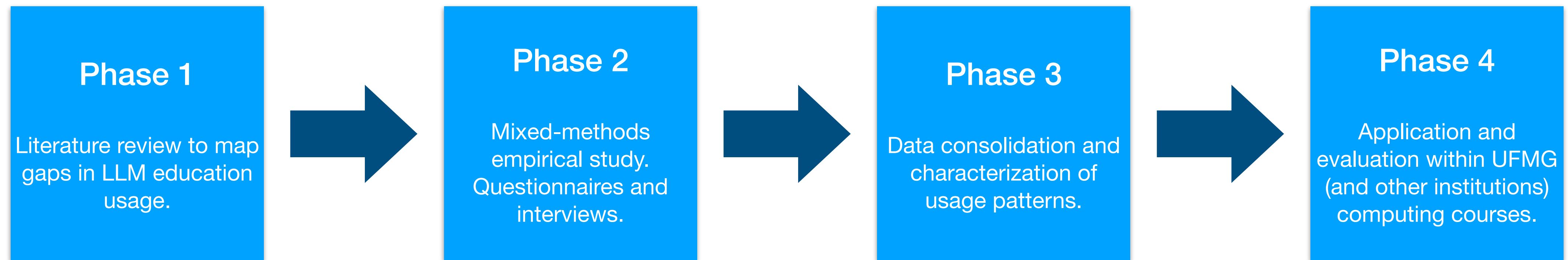
- Böttcher et al. [2025]
 - Recommended adapting student learning objectives and assessment methods for the AI era
- Haindl and Weinberger [2024]
 - Showed ChatGPT aids basic learning but struggles with complex concepts like object-oriented programming
- Barros et al. [2025]
 - Found that LLM-generated feedback effectively improves learning in introductory programming courses in Brazil

**Local gap:
Need to understand these dynamics
specifically in Brazil**

Research Proposal

- **In-depth investigation of how students in Brazilian higher education use LLMs for learning programming**
 - Diagnose and characterize the current situation in the country
 - Propose best practices for the use of GenAI programming
 - Suggest teaching methodologies and curricular adjustments
 - Identify the challenges in Brazilian specific context
- Related areas:
 - Software Engineering
 - Human-Computer Interaction: national and international HCI Grand Challenges agendas
 - GC6 [2025]: "Learning and Creativity"
 - GC4 BR [2024]: "Sociocultural Aspects in Human-Computer Interaction"

Study Design



Contributions

- Diagnosis
 - A clear characterization of LLM interaction in Brazilian institutions
- Best Practices
 - Practical guidelines for students to develop skills alongside AI
- Curricular Shift
 - Proposed adjustments for higher education programming syllabi
- HCI Challenges
 - Contributions to International and National Grand Challenges

Limitations

- Practical evaluation hurdles
 - Real-world testing depends on instructor availability and course plans
- Alternative scenarios
 - May require shifting evaluation to community workshops or extension courses if university courses are unavailable

Conclusion

- Curricular Adaptation
 - Higher education programs must urgently be updated to reflect the reality of AI-assisted programming and address instructor concerns over student reliance
- Importance of Local Context
 - Investigating the specific Brazilian educational landscape fills a major literature gap and prevents computational solutions from reinforcing discrimination and exclusion
- Advancing Grand Challenges
 - The research directly supports both international (Learning and Creativity) and national (Sociocultural Aspects) research agendas in Human-Computer Interaction
- Broad Scientific Value
 - The systematized knowledge will strongly contribute to the fields of Software Engineering and HCI, even if practical evaluation in a real course proves unfeasible

Thank you!

Questions?